Predicting Inferior Temporal Cortex Responses During Visual Object Recognition

EPFL NX-414 Mini Project
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I. Introduction

This mini-project aims to predict neural responses in the inferior temporal (IT) cortex of non-human primates to visual stimuli. Using the Majaj et al. (2015) dataset [1], which pairs object images with multi-unit neuron activity across IT sites, we compare model strategies from basic pixel-level regression to transfer learning from complex task-driven deep networks and end-to-end data-driven architectures. By evaluating each model's predictive performance, we seek to identify the framework that best captures IT cortex neural dynamics.

II. Dataset and Preprocessing

The dataset comprises 3,200 224×224 grayscale images of objects from 64 3D models on achromatic backgrounds with random pose, scale, and position. FFor each image, the mean firing rate of 168 IT multi-unit sites is recorded in the 70-170 ms post-stimulus window. For compatibility with ImageNet-pretrained CNNs, each image is duplicated across R, G, B channels, scaled to $[0, 1]$, and normalized with ImageNet's per-channel mean and std (*ImageNet compatible data*); for visualization or training from scratch, we denormalize back to $[0, 1]$ and keep a single channel (*processed data*). Data splits are 80% train, 20% validation (for tuning), and 20% test (for final evaluation).

III. Performance Metrics

For each of the 168 IT sites and each model, we compute two metrics over all stimuli to assess prediction performance: **Explained Variance (EV)** - the fraction of neural response variance captured by the model - and **Pearson Correlation** (r) - the strength of the linear relationship between actual and predicted responses. For model comparison, we summarize each metric's distribution across sites by its median and interquartile range : **median** (P_{75}, P_{25})

IV. Models Exploration

A. Baseline Linear Models

As a naïve baseline, we flatten each image of the *processed data* to a $224 \times 224 = 50176$ dim vector, standardize features, and fit OLS and Ridge ($\alpha = 10^3$) regressors, yielding median EV $= -0.437(-0.548\,, 0.301)$ vs. $-0.210(-0.319\,, -0.093)$ and median $r = 0.177(0.120\,, 0.257)$ vs. $0.207(0.126\,, 0.283)$. Both below EV=0, highlighting the difficulty of learning from high-dimensional data with $n \ll d$ without strong regularization. To adress this we explore PCA and conduct a 5-fold grid search over $k \in \{400, 600, \ldots, 1800\}$ and $\alpha \in 10^{0:5}$, the best configuration ($\alpha = 10^5,\ k = 1200$) achieves: median EV $= 0.075(0.0328\,, 0.124)$ and median $r = 0.278(0.188\,, 0.356)$.

B. Data-Driven Models

To explore the data driven approach we train, on the *processed data* a lightweight convolutional neural network (VanillaCNN) from scratch : a $1\times224\times224$ input passes through three 3×3 conv–BN–ReLU–2×2 max-pool blocks (channels $1 \to 8 \to 16 \to 32$, spatial $224 \to 112 \to 56 \to 28$), is flattened ($32 \times 28 \times 28 = 25\,088$), then fed to a 512-unit ReLU fully conected layer and projected to 168 outputs via a linear layer. We train the model for 50 epochs with mean squared error (MSE) loss and AdamW (learinng rate 10^{-3}, weight decay 0.01) achieving on the validation set : median EV $= 0.279\ (0.175,\ 0.351)$, and median $r = 0.538\ (0.432,\ 0.611)$.

C. Task-Driven Models

To explore the task-driven potential of deep features, we use the PyTorch implementation of a ResNet-50 pretrained on ImageNet [2] with frozen weights. For each stimulus in the *ImageNet-compatible data*, we extract activations from six intermediate layers:

$$\{\texttt{conv1}, \texttt{layer1}, \texttt{layer2}, \texttt{layer3}, \texttt{layer4}, \texttt{avgpool}\}.$$

Each layer's feature map is flatten and fed into our optimized linear-regression pipeline (PCA ($k = 1200$) $\to$ Ridge regression ($\alpha = 10^5$)). This fixed-pipeline ensures that any performance gain over the linear-baseline reflects pure representational improvement. The best validation performance is obtained from `layer4`: median EV $= 0.388\ (0.287,\ 0.483)$, median $r = 0.636\ (0.537,\ 0.700)$. As a control, we repeat the procedure on `layer4` using a randomly initialized ResNet-50, yielding significantly lower performance: median EV $= 0.121\ (0.001,\ 0.218)$, median $r = 0.405\ (0.286,\ 0.489)$, highlighting the benefits of task-driven pretrained representations. Figure 1 in Appendix shows model performance by layer depth, with deeper layers generally providing more predictive features. `Avgpool` underperforms, likely due to spatial compression and further dimensionality loss from PCA, limiting its ability to model 168 outputs

V. Most Effective Model

Based on our earlier findings, transfer learning via pretrained task-driven models emerged as the most effective approach. We therefore extracted features from the PyTorch implementation of ResNet-50's `layer3` and `layer4` and optimised two predictive pipelines on top of each: (i) PCA retaining 99% of the variance or no PCA followed by ridge regression with the penalty α tuned by 4-fold cross-validation; and (ii) a two-layer feedforward neural network trained on the PCA-reduced features. In both cases, `layer3` representations consistently outperformed those from `layer4`, and PCA yielded gains. As an example, PCA+ridge on `layer3` achieved median EV $= 0.4356\ (0.3234,\ 0.5301)$, median $r = 0.6710\ (0.5709,\ 0.7373)$,, versus median EV $= 0.3915\ (0.2877,\ 0.4879)$, median $r = 0.6398\ (0.5372,\ 0.7023)$. for `layer4`. Similar improvements of the same order of magnitude were observed with the FNN. After hyperparameter tuning, the best overall results were obtained with a FNN that consisted of two fully connected layers: an input layer mapping the PCA-reduced `layer3` features ($200704 \to 2425$ dimensions) to a 1 024-unit hidden layer (ReLU activation with 30% dropout), followed by an output layer projecting to 168 IT sites. This FNN was trained end-to-end for 400 epochs with mean squared error loss and the Adam optimizer (learning rate 5×10^{-5}, weight decay 5×10^{-4}), yielding the highest predictive performances overall : median EV $= 0.4658\ (0.3516,\ 0.5549)$, median $r = 0.6880\ (0.5941,\ 0.7504)$. Table I in Appendix shows performances of all our models.

VI. Appendix

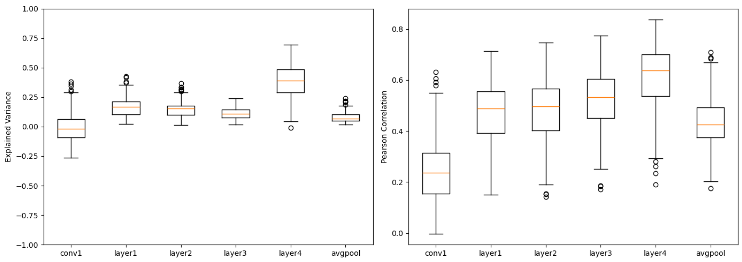

Fig. 1: Models performance as a function of pretrained ResNet50 layer depth

Model	Median EV (P_{25}, P_{75})	Median r (P_{25}, P_{75})
OLS Regressor	-0.437 (–0.548, 0.301)	0.177 (0.120, 0.257)
Ridge ($\alpha = 10^3$)	-0.210 (–0.319, –0.093)	0.207 (0.126, 0.283)
PCA + Ridge ($\alpha = 10^5$, $k = 1200$)	0.075 (0.0328, 0.124)	0.278 (0.188, 0.356)
ResNet-50 (random init, layer4)	0.121 (0.001, 0.218)	0.405 (0.286, 0.489)
VanillaCNN	0.279 (0.175, 0.351)	0.538 (0.432, 0.611)
ResNet-50 (pretrained, layer4)	0.388 (0.287, 0.483)	0.636 (0.537, 0.700)
Best Model	**0.465** (0.351, 0.554)	**0.688** (0.594, 0.750)

TABLE I: Performance of different models on the validation set, measured by median explained variance (EV) and median Pearson r with interquartile ranges (P_{25}–P_{75}).

References

[1] N. J. Majaj, H. Hong, E. A. Solomon, and J. J. DiCarlo, "Simple learned weighted sums of inferior temporal neuronal firing rates accurately predict human core object recognition performance," *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience*, vol. 35, p. 13402–13418, Sept. 2015.

[2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," Dec. 2015. arXiv:1512.03385 [cs].